

CO-1 ASSESSMENT TOOL-1

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Course

Code: ITA0612

course: Machine Learning

Slot: A

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Application-based questions

1. A factory has three boxes containing coloured balls. Box A contains 6 red, 4 blue and 2 green balls. Box B contains 5 red, 7 blue, and 3 green balls. and Box C contains 8 red, 2 blue, and 5 green balls. A ball drawn from the chosen box is found to be blue. what is the probability that the selected box was Box C?

Sol:

Given:

- Box A: 6 Red, 4 Blue, 2 Green $\rightarrow$ Total = 12
- Box B: 5 Red, 7 Blue, 3 Green $\rightarrow$ Total = 15
- Box C: 8 Red, 2 Blue, 5 Green $\rightarrow$ Total = 15

Since Box B is twice as likely as A and C:

$$P(A) = 1/4, P(B) = 1/2, P(C) = 1/4$$

Probability of drawing blue:

$$P(\text{Blue}|A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue}|B) = 7/15$$

$$P(\text{Blue}|C) = 2/15$$

Using Bayes' theorem:

$$P(C|\text{Blue}) = \frac{P(C)P(\text{Blue}|C)}{P(A)P(\text{Blue}|A) + P(B)P(\text{Blue}|B) + P(C)P(\text{Blue}|C)}$$

$$= \frac{1/4 \times 2/15}{\frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15}}$$

$$= \frac{1/30}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}}$$

$$L.C.M = 60:$$

$$= \frac{\frac{2}{60}}{\frac{5}{60} + \frac{14}{60} + \frac{2}{60}} = \frac{2}{21}$$

$$\therefore P(c|Blue) = \frac{2}{21} \approx 0.0952 \text{ (or) } 9.52\%$$

2) A Parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue and 5 black cars; and Section C has 7 red, 2 blue and 6 black cars. A randomly selected car is to blue. What is the Probability that the car came from Section C?

Given:

- Section A: 5 Red, 3 Blue, 2 Black $\rightarrow$ Total = 10
- Section B: 4 Red, 6 Blue, 5 Black $\rightarrow$ Total = 15
- Section C: 7 Red, 2 Blue, 6 Black $\rightarrow$ Total = 15

Section B is twice as likely:

$$P(A) = 1/4, P(B) = 1/2, P(C) = 1/4$$

Blue Probabilities:

$$P(Blue|A) = 3/10; P(Blue|B) = 6/15 = 2/5$$

$$P(Blue|c) = \frac{2}{15}$$

Using Bayes' theorem:

$$\begin{aligned} P(c|Blue) &= \frac{P(c)P(Blue|c)}{P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|c)} \\ &= \frac{1/4 \times 2/15}{\frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15}} \\ &= \frac{\frac{1}{30}}{\frac{3}{40} + \frac{1}{5} + \frac{1}{30}} \end{aligned}$$

$$L.C.M = 120:$$

$$= \frac{\frac{4}{120}}{\frac{9}{120} + \frac{24}{120} + \frac{4}{120}} = \frac{4}{37}$$

$$\therefore P(c|Blue) = \frac{4}{37} \approx 0.1081 \text{ (or) } 10.81\%$$

In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).

* Front as Front: 119 ; * Front as Back: 06

* Back as Back: 120 ; * Back as Front: 05.

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Given:

Actual/Predicted	Front	Back
Front	119	6
Back	5	120

Total:

$$TP + TN + FP + FN = 119 + 120 + 5 + 6 = 250.$$

$$1) \text{ Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{119 + 120}{250} = \frac{239}{250} = 95.6\%$$

$$\therefore \text{Accuracy} = 95.6\%$$

$$2) \text{ Precision} = \frac{TP}{TP + FP} = \frac{119}{119 + 5} = \frac{119}{124} = 95.97\%$$

$$\therefore \text{Precision} = 95.97\%$$

3) Sensitivity / Recall:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{119}{119 + 6} = \frac{119}{125}$$

$$\therefore \text{Sensitivity} = 95.2\%$$

4) Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{120}{120 + 5} = \frac{120}{125}$$

$$\therefore \text{Specificity} = 96\%$$

5) F1-Score

$$F_1 = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}}$$

$$F_1 = \frac{2(0.9597)(0.952)}{0.9597 + 0.952}$$

$$F_1 \approx 95.58\%$$

4. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy):

a) Diseased as Diseased: 138 b) Diseased as Healthy: 12

b) Healthy identified as Healthy: 135 d) Healthy as Diseased: 15

Compute: Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.

Given:

- Diseased $\rightarrow$ Diseased = 138 (TP)
- Diseased $\rightarrow$ Healthy = 12 (FN)
- Healthy $\rightarrow$ Healthy = 135 (TN)
- Healthy $\rightarrow$ Diseased = 15 (FP)

$$138 + 12 + 135 + 15 = 300$$

1. Accuracy:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}}$$

$$= \frac{138 + 135}{300} = \frac{273}{300}$$

$$\text{Accuracy} = 91\%$$

2. Precision:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$= \frac{138}{138 + 15} = \frac{138}{153}$$

$$\text{Precision} = 90.20\%$$

Sensitivity / Recall

$$\begin{aligned}\text{sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{138}{138 + 12} = \frac{138}{150}\end{aligned}$$

$$\boxed{\text{Sensitivity} = 92\%}$$

4. Specificity:

$$\begin{aligned}\text{Specificity} &= \frac{TN}{TN + FP} \\ &= \frac{135}{135 + 15} = \frac{135}{150}\end{aligned}$$

$$\boxed{\text{Specificity} = 90\%}$$

5. F1 - Score:

$$\begin{aligned}F1 &= \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}} \\ &= \frac{2(0.9020)(0.92)}{0.9020 + 0.92}\end{aligned}$$

$$\boxed{F1 \approx 91.00\%}$$

5. In a spam email filtering system, an AI model classifies emails as spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam):

- Spam as Spam: 180
- Spam as Not Spam: 20
- Not spam as Not Spam: 170
- Not spam as Spam: 30

Compute : Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Given:

- Spam $\rightarrow$ Spam = 180 (TP)
- Spam $\rightarrow$ Not spam = 20 (FN)
- Not spam $\rightarrow$ Not spam = 170 (TN)
- Not spam $\rightarrow$ Spam = 30 (FP)

Total:

$$180 + 20 + 170 + 30 = 400$$

1. Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

$$= \frac{180 + 170}{400} = \frac{350}{400}$$

$$\text{Accuracy} = 87.5\%$$

2. Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{180}{180 + 30} = \frac{180}{210}$$

$$\text{Precision} = 85.71\%$$

3. Sensitivity / Recall:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{180}{180 + 20} = \frac{180}{200}$$

$$\text{Sensitivity} = 90\%$$

4. Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{170}{170 + 30} = \frac{170}{200}$$

$$\text{Specificity} = 85\%$$

5. F1-Score

$$F1 = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}}$$

$$= \frac{2(0.8571)(0.90)}{0.8571 + 0.90}$$

$$F1 \approx 87.27\%$$